LIFELINE Iteration 1: Approach Note (Group 1)

Seed s = group number. Every random draw uses its own stream keyed by (s, case number, module), so one case never changes another.

1. From story to numbers

Delta Flood. E is drawn from 1.1V to 1.6V (a flat web, degree 2.2 to 3.2) and tree edges attach to one of the last six nodes, so villages form a wide strip. 45% or more of locations get penalty 2 or higher ("small villages under water"); 3 to 5 waves ("in waves"); crates are forced to 55% or more food and water, with a small margin.

Mountain Earthquake. E runs from V to 1.3V: a tree plus a few extra roads ("almost no alternatives"). Medicine and tools are forced to 45% or more. Alarm ticks come in bursts (70% of alarms cluster around random centres, "bursts follow each aftershock") and priorities are skewed high, so the mean is near 7.

Coastal Cyclone. At least 30% of locations are shelters. Windows are 120 to 480 minutes, started after minute 200, and redrawn until the mean overlap is 2.4 or more ("short windows that overlap" is met through high density, with the mean checked). The log is uniform over all 38 symbols plus repeated beacons ("full of noise"), 30 or more symbols checked.

Urban Flash Flood. E is 1.75V to 3V ("many alternative routes"). On M and L the alarm rate is 1.5 to 1.75 per tick ("faster than anyone can answer", above one release per tick). Rare crates are 10% plus one.

Landslide Corridor. A path 0..V-1 with chords (i, i+2) only: degree at most 4, E at most 1.2V, distance to the far end at least V/2. One path road in the middle, spanned by no chord, is forced into a wave, and a target is placed beyond it, so the waves cut it off.

Outbreak Camp. Medicine and water are forced to 55% or more; half of the crates plus one get expiry 0 to 300 ("supplies spoil"). One symbol has probability 0.34 ("beacons repeat"), checked at 27% or more; beacon strings are pasted into the log at random places.

Common choices: weights 1 to 20 with value near 1.5 to 6 times weight (rare crates +20); capacity 15% to 60% of total weight; windows always include three touching follow-up windows; every module size is drawn inside its tier range.

2. Planted objects and claims

Tour: tier S is the exact optimum by trying all (n-1)! orders, so tour_opt equals the planted tour. Other tiers use nearest neighbour then 2-opt, and the generator asserts it is not longer than the nearest neighbour tour under both tie rules (lowest and highest index). **Cover:** pick k nodes first, then create only edges with an end in the cover, joining the cover by a tree and attaching every other node to it; this gives at least n-1 edges and a valid cover by construction. **Safety:** draw the assignment first; each rule has three random literals and one sign is flipped if no literal holds. The rules per switch for M and L follow a shuffled list of 7 low (2.7 to 3.35), 11 middle (3.7 to 4.6) and 8 high (5.0 to 5.8) targets. **Rations:** choose a subset first and set the target to its sum. **Reports:** report_b comes from random edits of report_a and edit_bound is the exact edit distance. **Patterns:** four to five are substrings of the log, pattern 5 (and sometimes 4) is a random string re-drawn until absent; counts use overlapping search. **Claims (tier S):** knapsack DP, earliest finish greedy and weighted interval DP, rod cutting DP, tour by full search, and minimum cover by trying all 2^n subsets. **Variety:** in tier S cases with k mod 3 = 0 crates are redrawn until greedy by value per kg is not optimal (7 cases); in cases 0 to 3, 5 to 8 the graph is redrawn until a shelter or hospital has a fewest-roads cost above its cheapest cost (8 cases). Different cases come from different streams, and the validator also compares modules of size 8 or more for equality.

3. The traps

46 Dock order. 1,500 to 3,000 crates whose value, expiry and serial are sorted (three monotone fields; weight stays random). A sort with a first or last element pivot degrades to O(n^2) and the data hides that a loader must still decide on weight. Check V8 tests monotone fields in crate_id order. **47 Best-looking crate.** Twelve to fourteen crates: one 1 kg crate worth 15, two 10 kg crates worth 100, the rest 10 to 12 kg crates worth 10 to 25; capacity 20. Greedy takes the 15, then one 100, ends at 115; the optimum is 200 (57.5%). V8 reruns greedy and a DP. **48 Shortest looks cheapest.** A chain of 1,500 to 3,000 locations with chords (i, i+2) (distance to the far end about V/2, at least 500). Target 2 is reached directly with a 60 minute road plus penalty 6 (cost 66), or through location 1 for cost 8, a ratio above 8. V8 does a double sweep BFS for the length and a layered search for the cheapest fewest-roads cost. **49 Near miss.** A log of 50,000 to 100,000 symbols: one symbol repeated, with 3% to 7% noise that never contains that symbol or the mismatch symbol. Patterns 2 to 5 (lengths 8 to 40) are the repeated symbol m-1 times plus the mismatch symbol, so every window begins a near match (about 93% of positions) and a naive search does about N m steps. V8 counts block repetition and prefix matches.

4. What will be hard

Delta Flood. Route planning with closures (waves leave only long, penalised detours), and loading (55% food and water makes value per kg nearly equal, so greedy is wrong).

Mountain Earthquake. Alarm queue (bursts and a mean priority near 7 make deep backlogs), and route planning (few alternatives: one closure can disconnect a target, so the answer is often none).

Coastal Cyclone. Rescue scheduling (mean overlap of 2.4 or more makes the weighted choice non-greedy), and radio search and Huffman coding (all 38 symbols in noise give a nearly flat code).

Urban Flash Flood. Network planning and routing on dense graphs (degree 3.5 or more, many equal-cost routes), and the queue (rate 1.5 to 1.75 per tick means the backlog never clears).

Landslide Corridor. Reachability after waves (the cut leaves a target unreachable, so the code must return none and report the reachable set), and the spanning tree on a long thin graph with deep paths.

Outbreak Camp. Loading with spoilage-heavy medicine and water crates, and compression (one symbol at 27% or more gives a very short code for it and long ones for the rest).